

MS 6015: Multivariate Statistical Methods

Course Review

Reference: Johnson & Wichern, Applied Multivariate Statistical Analysis

2026

Course Structure

Module 1: Basics and Inference

- Matrix Algebra

- Multivariate Normal Distribution

- Multivariate Inference: Inference About the Mean

- Multivariate Inference: ANOVA and MANOVA

Module 2: Regression and Classification

- Simple Linear Regression

- Multiple Linear Regression

- Logistic Regression

- Feature Selection

- Classification

Module 3; Analysis of Interdependence

- PCA

- Factor Analysis

- Cluster Analysis

End Sem Exam

Why Geometry for Multivariate Data?

Let $\mathbf{X} \in \mathbb{R}^{n \times p}$ denote multivariate data.

- ▶ **Row geometry**: each observation is a point $\mathbf{x}_i \in \mathbb{R}^p$
- ▶ **Column geometry**: each variable is a vector in $\mathbb{R}^n$

Geometry underpins:

- ▶ Similarity, distance, and clustering
- ▶ Regression as orthogonal projection
- ▶ PCA as best low-dimensional representation

Vectors: Norms and Distance

A single variable across n observations is a column vector:

$$\mathbf{x} = (x_1, x_2, \dots, x_n)^\top \in \mathbb{R}^n, \quad \|\mathbf{x}\| = \sqrt{\mathbf{x}^\top \mathbf{x}}$$

Euclidean Distance

$d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|$ — scale of variables changes distances dramatically.

Dot Product and Angle

$\mathbf{x}^\top \mathbf{y} = \|\mathbf{x}\| \|\mathbf{y}\| \cos \theta$ — correlation is cosine similarity after centering and scaling.

- ▶ Sample variance: $s^2 = \frac{1}{n-1} \|\mathbf{x}_c\|^2$, where $\mathbf{x}_c = \mathbf{x} - \bar{x}\mathbf{1}$
- ▶ **Mahalanobis distance** rescales using the covariance matrix.

Matrices: Key Concepts

Eigenvalues & Eigenvectors

- ▶ $\mathbf{Ae} = \lambda \mathbf{e}$
- ▶ Describe stretching/rotation
- ▶ Foundation of PCA

Quadratic Forms

- ▶ $Q(\mathbf{x}) = \mathbf{x}^\top \mathbf{A} \mathbf{x}$
- ▶ Positive definite: $Q > 0$ for all $\mathbf{x} \neq \mathbf{0}$
- ▶ Covariance matrices are positive semi-definite

Random Vectors

- ▶ Mean vector: $\boldsymbol{\mu} = E(\mathbf{X})$
- ▶ Covariance matrix:
 $\boldsymbol{\Sigma} = E[(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^\top]$
- ▶ Diagonal: variances;
off-diagonal: covariances

SVD & Rotation

- ▶ Any matrix: $\mathbf{A} = \mathbf{U} \mathbf{D} \mathbf{V}^\top$
- ▶ Orthogonal matrices rotate vectors
- ▶ Connects to PCA geometry

The Multivariate Normal Distribution

$\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ has density:

$$f(\mathbf{x}) = \frac{1}{(2\pi)^{p/2} |\boldsymbol{\Sigma}|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right)$$

Key Properties

- ▶ Any subset of variables is also multivariate normal
- ▶ Any linear combination $\mathbf{a}^\top \mathbf{X} \sim N(\mathbf{a}^\top \boldsymbol{\mu}, \mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a})$
- ▶ Zero covariance $\Leftrightarrow$ independence (unique to MVN)
- ▶ Conditional distributions are also normal:
 $\mathbf{X}_1 | \mathbf{X}_2 = \mathbf{x}_2 \sim N(\boldsymbol{\mu}_{1|2}, \boldsymbol{\Sigma}_{11|2})$

Probability contours are ellipses determined by $\boldsymbol{\Sigma}^{-1}$.

Sampling Distributions of $\bar{\mathbf{X}}$ and $\mathbf{S}$

Given a random sample $\mathbf{x}_1, \dots, \mathbf{x}_n \stackrel{iid}{\sim} N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$:

- ▶ $\bar{\mathbf{X}} \sim N_p(\boldsymbol{\mu}, \frac{1}{n}\boldsymbol{\Sigma})$
- ▶ $(n-1)\mathbf{S} \sim W_p(n-1, \boldsymbol{\Sigma})$ (Wishart distribution)
- ▶ $\bar{\mathbf{X}}$ and $\mathbf{S}$ are **independent**

Central Limit Theorem (Multivariate)

For large n , $\sqrt{n}(\bar{\mathbf{X}} - \boldsymbol{\mu}) \xrightarrow{d} N_p(\mathbf{0}, \boldsymbol{\Sigma})$ regardless of the population distribution.

From t -test to Hotelling's T^2

Univariate	Multivariate
$H_0 : \mu = \mu_0$	$H_0 : \boldsymbol{\mu} = \boldsymbol{\mu}_0$
$t^2 = n(\bar{x} - \mu_0)^\top s^{-2}(\bar{x} - \mu_0)$	$T^2 = n(\bar{\mathbf{x}} - \boldsymbol{\mu}_0)^\top \mathbf{S}^{-1}(\bar{\mathbf{x}} - \boldsymbol{\mu}_0)$
$t^2 \sim F_{1, n-1}$	$\frac{n-p}{(n-1)p} T^2 \sim F_{p, n-p}$

- ▶ T^2 is **invariant** to linear transformations $\mathbf{Y} = \mathbf{C}\mathbf{X} + \mathbf{d}$
- ▶ **Wilks' Lambda**: $\Lambda^{2/n} = |\hat{\boldsymbol{\Sigma}}|/|\hat{\boldsymbol{\Sigma}}_0|$ (likelihood ratio)

Confidence Regions and Simultaneous CIs

Hotelling Ellipsoidal Confidence Region

A $(1 - \alpha)$ confidence region for μ :

$$\left\{ \mu : n(\bar{\mathbf{x}} - \mu)^\top \mathbf{S}^{-1}(\bar{\mathbf{x}} - \mu) \leq \frac{p(n-1)}{n-p} F_{p, n-p}(\alpha) \right\}$$

Three approaches to simultaneous inference:

- ▶ **One-at-a-time** (marginal) CIs — ignores multiple testing
- ▶ **Bonferroni** CIs — conservative correction, α/p per interval
- ▶ **T^2 simultaneous CIs** — exact coverage for all linear combinations

ANOVA: One-Way

Goal: Compare means $\mu_1, \dots, \mu_g$ of g groups on one response.

$$H_0 : \mu_1 = \mu_2 = \dots = \mu_g$$

Partition total variation:

$$SS_{\text{Total}} = SS_{\text{Treatment}} + SS_{\text{Error}}$$

- ▶ Test statistic: $F = \frac{MS_{\text{Treatment}}}{MS_{\text{Error}}} \sim F_{g-1, N-g}$
- ▶ Assumptions: normality, homogeneity of variance, independence
- ▶ Two-way ANOVA adds a blocking or second factor, partitioning: $SS_T = SS_{\text{Treatment}} + SS_{\text{Block}} + SS_{\text{Error}}$

MANOVA: Multivariate Extension

Goal: Compare mean *vectors* across g groups on p responses simultaneously.

$$H_0 : \mu_1 = \mu_2 = \cdots = \mu_g$$

SSCP matrices replace sums of squares:

$$\mathbf{T} = \mathbf{H} + \mathbf{W} \quad (\text{Total} = \text{Hypothesis} + \text{Error})$$

Wilks' Lambda Test Statistic

$$\Lambda = \frac{|\mathbf{W}|}{|\mathbf{H} + \mathbf{W}|} \quad (\text{small } \Lambda \Rightarrow \text{reject } H_0)$$

Other test criteria: Pillai's trace, Lawley-Hotelling trace, Roy's largest root. Two-way MANOVA extends to block designs:

$$\mathbf{T} = \mathbf{H} + \mathbf{B} + \mathbf{W}.$$

ANOVA vs. MANOVA: Core Comparison

	ANOVA	MANOVA
Responses	One variable	Several variables
Mean comparison	Scalar μ	Vector $\boldsymbol{\mu}$
Test statistic	F -ratio	Wilks' Λ
Advantage	Simplicity	Guards against Type I error across multiple tests

MANOVA detects group differences that ANOVA on each response separately may miss.

Simple Linear Regression (SLR)

$$Y = \beta_0 + \beta_1 X + \varepsilon, \quad \varepsilon \sim N(0, \sigma^2)$$

- ▶ β_0 : intercept (mean of Y when $X = 0$)
- ▶ β_1 : slope (change in mean of Y per unit increase in X)
- ▶ Estimated by **OLS**: minimise $\sum (y_i - \hat{y}_i)^2$

Inference on β_1

$$s_{b_1}^2 = \frac{s^2}{(n-1)s_X^2}, \quad t = \frac{b_1 - \beta_1^0}{s_{b_1}} \sim t_{n-2}$$

- ▶ Model fit: $R^2 = 1 - SS_{\text{Res}}/SS_{\text{Total}}$

Multiple Linear Regression (MLR)

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim N(\mathbf{0}, \sigma^2 \mathbf{I})$$

$$\text{OLS estimator: } \hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

Testing individual slopes

$$t_j = \frac{b_j}{s_{b_j}} \sim t_{n-k-1}$$

Overall F -test

$$F = \frac{MS_{\text{Reg}}}{MS_{\text{Res}}} \sim F_{k, n-k-1}$$

Partial F -test

Compare full vs. reduced model:

$$F = \frac{(SS_{\text{Res,red}} - SS_{\text{Res,full}})/q}{MS_{\text{Res,full}}}$$

Tests whether a *subset* of q predictors contribute.

Multicollinearity

Problem: When predictors are highly correlated, standard errors inflate and slopes become unstable.

Variance Inflation Factor (VIF)

$$\text{VIF}(X_j) = \frac{1}{1 - R_j^2}, \quad \text{se}(b_j) \propto \sqrt{\text{VIF}(X_j)}$$

where R_j^2 is the R^2 from regressing X_j on all other predictors.

Rules of thumb: $\text{VIF} > 5$ — moderate concern; $\text{VIF} > 10$ — serious. **Warning signs:**

- ▶ High F , but weak individual t -statistics
- ▶ Slopes change dramatically when a variable is added/removed
- ▶ R^2 increases less than expected when adding a predictor

Remedies: remove redundant predictors, ridge regression, PCA-based regression.

Analysis of Covariance (ANCOVA)

Purpose: Compare group means while *adjusting* for one or more continuous confounding variables (covariates).

$$Y = \beta_0 + \beta_1 X_{\text{covariate}} + \beta_2 \cdot \text{Group} + \varepsilon$$

- ▶ Combines ANOVA (categorical predictor) with regression (continuous covariate)
- ▶ The dummy variable coefficient β_2 = adjusted difference between group intercepts
- ▶ An **interaction term** $\text{Group} \times \text{Covariate}$ tests whether slopes differ across groups

Key condition: parallel slopes (no interaction) for standard ANCOVA.

Logistic Regression

- ▶ The logit is a function of the probability p . The logit graphs as a straight line in the predictor variable X so

$$\text{logit}(p) = \ln \left(\frac{p}{1-p} \right) = \beta_0 + \beta_1 X$$

- ▶ Regression parameters in the case of logistic regression are estimated using Maximum Likelihood Estimator (MLE).
- ▶ β_1 is the change in the log-natural of the ratio of odds. Alternatively, $\exp(\beta_1)$ or is the change in the odds ratio for unit change in the feature value

Model Diagnostics

- ▶ Omnibus test: Omnibus tests are generic class of tests in statistics which check whether the explained variance in the model is significantly higher than the unexplained variation.
- ▶ Wald's test: Wald's test is used for checking statistical significance of individual predictor variables (equivalent to t-test in MLR model).
- ▶ Analysis of Deviance

Feature (Variable) Selection

Goal: Find the best subset of predictors — balance fit and parsimony.

Algorithmic Methods

- ▶ **Forward selection:** add variables one at a time (LR or Wald test)
- ▶ **Backward elimination:** start full, remove weakest
- ▶ **Stepwise:** combination of forward/backward
- ▶ In logistic regression: Forward LR (likelihood ratio) vs. Forward Wald
- ▶ Over-fitting risk: always validate on hold-out data

Criterion-Based

- ▶ **AIC:** $-2\ell + 2k$
- ▶ **BIC:** $-2\ell + k \log n$
(penalises complexity more)
- ▶ **Adjusted R^2 :** penalises for extra parameters
- ▶ **Mallows' C_p**

Logistic Regression as a Classifier

- ▶ Classification table
- ▶ Classification metrics
 - ▶ **Accuracy**: Measures overall correctness.
 - ▶ **Precision**: Indicates accuracy of positive predictions.
 - ▶ **Recall (Sensitivity)**: Measures the proportion of actual positives correctly identified.
 - ▶ **Specificity**: Measures the proportion of actual negatives correctly identified.

Metrics	Values
Sensitivity (Recall)	$\frac{TP}{TP+FN}$
Specificity	$\frac{TN}{TN+FP}$
Accuracy	$\frac{TP+TN}{TP+TN+FP+FN}$
Precision	$\frac{TP}{TP+FP}$

- ▶ ROC, AUC, and choosing optimal cut-off

Classification Trees

- ▶ Classification models- CHAID
- ▶ Classification models- CART

Principal Component Analysis (PCA)

Goal: Reduce p correlated variables to $q \ll p$ uncorrelated components that capture maximum variance.

Definition

The i -th PC: $Y_i = \mathbf{e}_i^\top \mathbf{X}$, where $(\lambda_i, \mathbf{e}_i)$ is the i -th eigenvalue-eigenvector pair of $\mathbf{\Sigma}$.

$$\text{Var}(Y_i) = \lambda_i, \quad \text{Cov}(Y_i, Y_k) = 0 \text{ for } i \neq k$$

Proportion of total variance explained by first q PCs:

$$\frac{\sum_{i=1}^q \lambda_i}{\sum_{i=1}^p \lambda_i}$$

Component loadings: $\rho_{Y_i, X_k} = e_{ik} \sqrt{\lambda_i} / \sqrt{\sigma_{kk}}$

Selecting q : scree plot (elbow), proportion of variance ≥ 70 –80%, eigenvalue > 1 (correlation PCA).

PCA: Interpretation via Loadings and Biplots

Loading matrix (example: 3 variables):

	Z_1	Z_2	Z_3
X_1	0.928	-0.080	-0.364
X_2	0.726	0.670	0.159
X_3	0.822	-0.501	0.271
% Var	68.8%	23.5%	7.7%

Z_1 captures a general factor;

X_1 – X_3 all load highly.

Biplot

- Overlays *scores* (observations in PC space) and *loading vectors* (arrows = correlations with PCs)
- Long arrow → variable well-represented
- Arrows close together → correlated variables

PCA vs. FA

PCA is descriptive (mathematical).
Factor Analysis posits a *causal latent structure*.

The Orthogonal Factor Model

Idea: Observable $\mathbf{X}$ is driven by a small number of unobservable *common factors* $F_1, \dots, F_m$:

$$X_i - \mu_i = l_{i1}F_1 + l_{i2}F_2 + \dots + l_{im}F_m + \varepsilon_i$$

- ▶ l_{ij} : **factor loadings** (regression weights on factors)
- ▶ ε_i : specific (unique) factor, $\text{Cov}(\mathbf{F}, \boldsymbol{\varepsilon}) = \mathbf{0}$
- ▶ Implied covariance: $\boldsymbol{\Sigma} = \mathbf{LL}^\top + \boldsymbol{\Psi}$

Communality and Specific Variance

$$\underbrace{\sigma_{ii}}_{\text{Var}(X_i)} = \underbrace{l_{i1}^2 + l_{i2}^2 + \dots + l_{im}^2}_{h_i^2 \text{ (communality)}} + \underbrace{\psi_i}_{\text{specific variance}}$$

Estimation and Rotation in Factor Analysis

Estimation Methods

- ▶ **Principal Component Method**: use first m eigenvectors of $\mathbf{S}$; no normality required
- ▶ **Maximum Likelihood**: assumes MVN; allows formal tests for number of factors

Factor Rotation

- ▶ Loadings are *indeterminate* up to an orthogonal matrix $\mathbf{T}$
- ▶ **Varimax** (orthogonal): maximises variance of squared loadings per factor → *simple structure*
- ▶ **Oblique** rotations: allow correlated factors

Factor Scores

Once factors are identified, *factor scores* assign each observation a position on the latent dimensions — useful for further modelling.

Number of factors: Scree plot, proportion of variance, likelihood ratio test.

Introduction to Cluster Analysis

Definition

Cluster analysis groups objects so that members of the same cluster are more similar to each other than to objects in other groups — *without prior labels*.

Types of Clustering

- ▶ **Non-overlapping**: each observation in exactly one cluster (most common)
- ▶ **Overlapping**: observation may belong to multiple clusters
- ▶ **Probabilistic**: soft membership probabilities (mixture models)

Clustering vs. Classification

- ▶ Clustering: *unsupervised* — classes not known a priori
- ▶ Classification: *supervised* — training labels available
- ▶ Clustering → descriptive analytics
- ▶ Classification → predictive analytics

Distance and Similarity Measures

Every clustering algorithm operates on a pairwise **distance matrix**.

Measure	Formula	Use case
Euclidean	$\ \mathbf{x} - \mathbf{y}\ $	Interval/ratio variable
Standardised Euclidean	Euclidean on z-scores	Different scales
Mahalanobis	$\sqrt{(\mathbf{x} - \mathbf{y})^\top \mathbf{S}^{-1}(\mathbf{x} - \mathbf{y})}$	Correlated variables
Manhattan	$\sum x_k - y_k $	Robust to outliers
Minkowski	$(\sum x_k - y_k ^m)^{1/m}$	Generalised
Jaccard	$(a + d)/(p)$ table	Binary variables

- ▶ For quantitative *and* qualitative variables: Gower's distance or cosine similarity
- ▶ Metric axioms: non-negativity, symmetry, triangle inequality

Hierarchical Clustering

Builds a **dendrogram** by successively merging (agglomerative) or splitting (divisive) clusters. **Linkage methods** (define inter-cluster distance):

Method	Distance between clusters
Single linkage	Minimum pairwise distance (chaining tendency)
Complete linkage	Maximum pairwise distance (compact clusters)
Average linkage	Mean of all pairwise distances (compromise)
Ward's method	Minimise within-cluster variance (most popular)

- ▶ Cut dendrogram at chosen height to get k clusters
- ▶ No need to pre-specify k ; exploratory
- ▶ Sensitive to outliers (especially single linkage)

Non-Hierarchical Clustering

k -Means Algorithm

1. Specify k ; initialise k cluster centres (centroids)
2. **Assignment**: assign each point to nearest centroid
3. **Update**: recompute centroids as cluster means
4. Repeat 2–3 until centroids do not change

Advantages

- ▶ Fast and scalable
- ▶ Simple to implement
- ▶ Works well for spherical, similarly-sized clusters

Limitations

- ▶ Must pre-specify k
- ▶ Hard (crisp) assignments only
- ▶ Sensitive to initialisation and outliers
- ▶ Assumes roughly equal cluster sizes and spherical shape

Clustering Based on Statistical Methods (GMM)

Gaussian Mixture Model

$$p(\mathbf{x}) = \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$$

Parameters estimated via the **EM algorithm** (E-step: responsibilities; M-step: parameter updates).

Advantages over *k*-means

- ▶ **Soft** (probabilistic) membership
- ▶ Elliptical cluster shapes
- ▶ Captures uncertainty
- ▶ Model selection via AIC/BIC

Course Road Map

Part	Topics
Introduction	Matrix algebra, vectors, geometry Multivariate normal distribution Sampling distributions
Multivariate Inference	One-sample Hotelling's T^2 Confidence regions, simultaneous CIs One-way and two-way ANOVA One-way and two-way MANOVA
Regression and Classification	Simple and multiple linear regression Multicollinearity (VIF) and ANCOVA Feature selection Logistic regression and Classification trees
Analysis of Interdependence	Principal Component Analysis Factor Analysis Cluster Analysis

Connecting Themes

- ▶ **Distance**: Euclidean $\rightarrow$ Mahalanobis $\rightarrow$ drives clustering, T^2 , and PCA
- ▶ **Eigendecomposition**: runs through covariance matrices, PCA, factor analysis, MANOVA
- ▶ **Univariate $\rightarrow$ Multivariate**: every univariate concept has a multivariate analogue ($t \rightarrow T^2$; ANOVA $\rightarrow$ MANOVA; variance $\rightarrow$ covariance matrix; $R^2 \rightarrow$ Wilks' Λ)
- ▶ **Supervised vs. Unsupervised**: regression/ANOVA use a response variable; PCA/FA/clustering do not
- ▶ **Dimension reduction**: PCA (descriptive), FA (latent structure), cluster analysis (prototype discovery)

Exam details

- ▶ Date of exam; May 14, 2026- Please check moodle for time and venue.
- ▶ Syllabus: Entire syllabus
- ▶ Marks: 40
- ▶ Breakup
 - ▶ Module 1: 10 marks
 - ▶ Module 2: 10 marks
 - ▶ Module 3: 20 marks
- ▶ Pattern: Mix of objective and short answer questions. It will be a pen-paper exam.